

Exchange Concepts Knowledge Check — Variant 02

Purpose: verify that the student has read [How a Financial Exchange Works](#) and internalized the core principles of a modern exchange, across its full scope — market history, order mechanics, risk and compliance, and technology infrastructure.

Name/ID

Instructions

- There are 20 questions.
- Each question has five statements (A-E) which may be false or true.
- Select all statements you believe to be correct by clearly filling the square before the question.
- Each question may have multiple correct statements (for some, all five statements may be correct).
- There are always at least one correct statement.

Scoring

- +1 point for each correct option selected.
- -2 points for each incorrect option selected.
- 0 points for options not selected.
- Final test score is capped at a minimum of 0: if raw total is negative, recorded score is 0.
- A passing score is 70% of the maximum rounded down to the nearest integer.

Questions

1. Pre-trade risk controls

- ☐ A. Pre-trade checks sit between the participant and the matching engine, so the engine itself can remain optimized purely for speed
- ☐ B. Position limits and credit limits measure exactly the same thing under different names
- ☐ C. Fail-fast check ordering typically evaluates cheap checks like format/syntax before expensive ones like live position/credit limits
- ☐ D. Rate limiting/throttling caps the number of orders a gateway can submit per second
- ☐ E. Position and credit-limit checks are usually the cheapest checks to perform, since they don't need external data

2. Circuit breakers, general

- ☐ A. Circuit breakers trace their origin to the aftermath of the 1987 Black Monday crash and the Brady Commission's recommendations
- ☐ B. During a circuit-breaker halt, new orders cannot be submitted at all until the halt ends
- ☐ C. A halt typically ends via an instant resumption of continuous matching, with no auction step
- ☐ D. Level 3 of the US market-wide circuit breaker halts trading for the remainder of the trading day
- ☐ E. Circuit-breaker thresholds and halt durations have never been adjusted since their original 1988 introduction

3. LULD and the COVID-19 circuit-breaker events

- ☐ A. Level 1 US market-wide circuit breakers were triggered four times during the March 2020 COVID-19 crash
- ☐ B. Level 2 was also triggered multiple times alongside Level 1 in March 2020
- ☐ C. Single-stock LULD pauses operate independently from market-wide circuit breakers and were triggered far more frequently in March 2020
- ☐ D. A stock that moves outside its LULD band and stays there beyond the monitoring window enters a fixed-duration trading pause
- ☐ E. Market-wide circuit-breaker reference levels reset after a Level 1 halt and resumption

4. The Mizuho fat-finger incident

- ☐ A. The 2005 Mizuho Securities incident involved a trader accidentally selling a huge quantity of shares at an absurdly low price instead of a small quantity at the intended price
- ☐ B. The Mizuho incident was caused by a matching-engine bug rather than a human order-entry error
- ☐ C. The Mizuho incident had no lasting influence on exchange risk-control practices
- ☐ D. Static price collars compare an order's price to the most recently traded price rather than the prior close
- ☐ E. The Mizuho incident occurred in the United States

5. Technology architecture

- ☐ A. The gateway is typically the participant-facing component responsible for authentication and session management
- ☐ B. FIX is a text-based protocol whose wire delimiter is a non-printable SOH character, not a literal pipe symbol
- ☐ C. Binary protocols likeITCH and OUCH are generally more compact and faster to parse than an equivalent FIX message
- ☐ D. Market data is commonly distributed over UDP multicast, while order submission commonly uses TCP for reliability
- ☐ E. A hybrid design using UDP multicast for live data plus a separate TCP unicast channel for gap recovery is a standard exchange market-data pattern

6. Conformance testing and onboarding

- ☐ A. Participants can typically connect a production gateway before completing formal conformance testing
- ☐ B. Conformance test scripts commonly simulate a disconnect/reconnect scenario to verify correct sequence-gap recovery
- ☐ C. A stale, out-of-sync certification/UAT environment is still considered a valid basis for certifying participants
- ☐ D. Recertification is typically required when the exchange changes something participants depend on, like a modified FIX tag
- ☐ E. Offboarding typically includes cancelling all resting orders and confirming final position/clearing reconciliation

7. Recovery objectives

- ☐ A. RPO measures the maximum acceptable downtime, while RTO measures the maximum acceptable data loss
- ☐ B. For a matching engine, RPO is effectively zero — no committed trade or acknowledgement can be lost
- ☐ C. Modern exchanges typically target sub-minute RTO for their most critical components
- ☐ D. The 2015 NYSE trading halt, lasting roughly 3.5 hours, is generally viewed as an example of an acceptable, well-managed RTO
- ☐ E. RTO tolerances are identical for a matching engine and for a batch reporting system

8. Failover and split-brain

- ☐ A. Split-brain occurs when both the primary and secondary site believe the other has failed, and each tries to become primary
- ☐ B. Fencing (STONITH) is a technique used to force a suspected-failed node to stop before a secondary takes over
- ☐ C. Active-active failover, where both sites simultaneously process live orders with full consensus, is the most common production pattern among exchanges
- ☐ D. Consensus protocols like Raft or Paxos can guarantee zero data loss on failover, at the cost of added latency
- ☐ E. Manual, operator-controlled failover is generally slower than automatic failover but reduces the risk of a false failover

9. Unique order numbering

- ☐ A. UUIDs guarantee practical global uniqueness without coordination, but they are not inherently sequential in arrival order
- ☐ B. A single central counter for order IDs has no drawbacks as a scaling solution
- ☐ C. Pre-allocated ID ranges per site can never create any gaps in the ID sequence
- ☐ D. Site-prefixed order IDs make it trivial to compare arrival order without stripping the prefix first
- ☐ E. Partitioning order books by symbol requires constant cross-server coordination for every single order

10. Market-data feed design

- ☐ A. A snapshot represents the complete current book state, while an incremental update represents only what changed
- ☐ B. Conflation merges consecutive updates to the same price level into a single message, which can lose intermediate states
- ☐ C. Conflated feeds are considered fully acceptable for systems that need complete tick-by-tick analytical data
- ☐ D. Sequence numbers allow a subscriber to detect that a message was missed between two consecutive numbers
- ☐ E. Top-of-book and depth-of-book data both convey identical bandwidth and processing requirements

11. SIP vs proprietary feeds

- ☐ A. The SIP is a public, consolidated feed aggregating data from every registered exchange under SEC oversight
- ☐ B. Proprietary direct feeds are generated straight from a venue's own matching engine, avoiding the SIP's aggregation step
- ☐ C. Proprietary feeds can offer both more information and lower latency compared to the SIP
- ☐ D. Exchanges that jointly govern the SIP's pricing are, at the same time, the same companies selling competing proprietary feeds — a structural conflict of interest critics have raised
- ☐ E. Market data and connectivity revenue has become a substantial and growing share of major exchange profit

12. PFOF and dark pool settlements

- ☐ A. Payment for order flow is banned in the UK and EU under MiFID II's best-execution requirements
- ☐ B. In the US, PFOF is fully banned outright with no ambiguity
- ☐ C. Barclays paid a settlement over allegations it misled clients about the presence of HFT activity in its dark pool
- ☐ D. Wholesale market makers pay for retail order flow because retail flow carries high adverse-selection risk
- ☐ E. Zero-commission retail trading has no meaningful connection to payment for order flow

13. Smart order routing and fragmentation

- ☐ A. The NBBO is computed in real time as the best available bid and best available ask across all exchanges
- ☐ B. A marketable order must be routed to satisfy the NBBO or better, not a worse displayed price
- ☐ C. A locked market occurs when the best bid on one venue equals the best ask on another venue
- ☐ D. A crossed market is actually less unusual and less exploitable than a locked market
- ☐ E. Smart order routers can consider factors like fees, queue-position probability, and venue toxicity signals, not just price alone

14. Execution algorithms

- ☐ A. A TWAP algorithm slices an order evenly across time regardless of actual traded volume, unlike VWAP, which weights toward higher-volume periods
- ☐ B. Implementation Shortfall algorithms use a fixed, unchanging schedule regardless of how price moves
- ☐ C. Percentage of Volume strategies aim to match a fixed absolute share count regardless of market volume
- ☐ D. VWAP algorithms ignore expected intraday volume patterns entirely
- ☐ E. TWAP and VWAP are two names for exactly the same execution strategy

15. Crypto venues

- ☐ A. Centralized crypto exchanges typically decouple trade matching from blockchain settlement, recording trades as internal ledger entries
- ☐ B. Decentralized exchanges commonly replace the order book with an automated market maker governed by a pricing formula like a constant-product rule
- ☐ C. A trade on a typical AMM-based DEX is its own settlement, executed atomically within a single blockchain transaction
- ☐ D. Cryptocurrency venues are subject to one single unified global regulator, unlike traditional securities markets
- ☐ E. Centralized crypto exchanges always operate with an independent, regulated central counterparty separate from the exchange itself

16. The FTX collapse

- ☐ A. FTX simultaneously operated as the trading venue and, through an affiliated firm, traded against its own customers
- ☐ B. FTX had an independent CCP performing novation between its customers, similar to traditional cleared markets
- ☐ C. Customer deposits were allegedly commingled with the affiliated trading firm's funds
- ☐ D. FTX's collapse has been compared to the Barings Bank and Knight Capital episodes as a cautionary tale about separation of duties
- ☐ E. FTX's founder was later convicted of fraud following the collapse

17. Latency and co-location

- ☐ A. Light travels faster through fibre-optic glass than radio waves travel through open air over the same distance
- ☐ B. Co-locating a participant's servers inside the exchange's own data center can reduce round-trip latency to low single-digit microseconds
- ☐ C. Spread Networks' expensive new fiber route became less competitive within a few years, once microwave links achieved similar routes faster
- ☐ D. FPGAs typically take milliseconds to parse a market-data message and generate a response
- ☐ E. PTP clock synchronization is generally less precise than NTP

18. Observability and the four golden signals

- ☐ A. The four golden signals used in SRE-style monitoring are Latency, Traffic, Errors, and Saturation
- ☐ B. Saturation is typically described as a leading indicator that tends to rise before latency and errors visibly degrade
- ☐ C. Metrics alone are generally sufficient to explain exactly why one specific order behaved oddly
- ☐ D. An error budget represents the gap between a service's SLO target and 100% availability/performance
- ☐ E. Symptom-based alerting and cause-based alerting are considered equally effective and interchangeable approaches

19. Corporate actions and reference data

- ☐ A. A stock split leaves total market capitalization unchanged, even though the per-share price and share count both change
- ☐ B. Open limit orders generally need adjustment across a stock split, so that price and quantity reflect the new terms
- ☐ C. A wrongly loaded contract multiplier can make every P&L and margin calculation for that instrument wrong by the same proportional factor
- ☐ D. Reference data is typically loaded and cached aggressively at startup, because querying it per order would add unacceptable latency
- ☐ E. Corporate actions require coordinated updates across the order book engine, clearing system, market data system, and audit trail

20. The 2015 NYSE halt

- ☐ A. The 2015 NYSE trading halt, lasting roughly 3.5 hours, was ultimately traced to a software configuration mismatch from a gateway update
 - ☐ B. During the 2015 NYSE halt, all US equity trading nationwide came to a complete stop because no other venue could handle NYSE-listed stocks
 - ☐ C. The 2015 NYSE halt was caused by a cyberattack on the exchange's matching engine
 - ☐ D. The 2015 NYSE halt demonstrated that reference-data/configuration issues cannot meaningfully affect exchange availability
 - ☐ E. The 2015 NYSE halt was resolved within a few seconds, due to automatic failover
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